

JUNO TH TRACKER TECHNICAL DATA SHEET EN



The Juno Tracker is a versatile tracker that uses mioty®, LoRaWAN® or cellular technologies such as NB-IoT and LTE-CAT-M1 for reliable and energy-efficient data transmission. The tracker is also available in different versions, e.g. for cold chain monitoring. Thanks to its robust design and IP69k/IP67 rating, it is ideal for use in demanding environments and can withstand dust, water and high pressures.

VERSION HISTORY

Version	Date	Revision
1.0.0	18.02.2025	Creation
1.0.1	24.04.2026	Formatting, version history

VERSIONS

Article code	Features
S-JUNO-LOEU-TRACK	JUNO Tracker with tilt and activity detection and opening detection LoRaWAN®, GNSS Scan, WIFI SSID Scan
S-JUNO-MIOTY-TRACK	JUNO Tracker with tilt and activity detection and opening detection LoRaWAN, GNSS Scan, WIFI SSID Scan mioty®
S-JUNO-IX-LOEU-TRACK	INDUSTRIAL JUNO Tracker with tilt and activity detection and opening detection LoRaWAN®, GNSS Scan, WIFI SSID Scan
S-JUNO-IX-MIOTY-TRACK	INDUSTRIAL JUNO Tracker with tilt and activity detection and mioty® opening detection, GNSS Scan, WIFI SSID Scan
S-JUNO-LOEU-TH-TRACK	JUNO IP67 TH Tracker, temperature and relative humidity with tilt and activity detection and LoRaWAN opening detection®
S-JUNO-MIOTY-TH-TRACK	JUNO IP67 TH Tracker, temperature and relative humidity with tilt and activity detection and opening detection mioty®
S-JUNO-IX-LOEU-TH-TRACK	INDUSTRIAL JUNO IP67 TH Tracker, temperature and relative humidity with tilt and activity detection and LoRaWAN opening detection®

S-JUNO-IX-MIOTY-TH-TRACK	INDUSTRIAL JUNO IP67 TH Tracker, temperature and relative humidity with tilt and activity detection and opening detection mioty®
S-JUNO-NBM1-TRACK-2	JUNO Tracker with tilt and activity detection and opening detection NB-IoT, LTE-CAT-M1 , GNSS, WIFI SSID Scan, with SIM card slot, two batteries
S-JUNO-NBM1-TRACK-3	JUNO Tracker with tilt and activity detection and opening detection NB-IoT, LTE-CAT-M1 , GNSS, WIFI SSID Scan, with on board SIM, three batteries
S-JUNO-IX-NBM1-TRACK-2	INDUSTRIAL JUNO tracker with tilt and activity detection and opening detection NB-IoT, LTE-CAT-M1 , GNSS, WIFI SSID scan, with SIM card slot, two batteries
S-JUNO-IX-NBM1-TRACK-3	INDUSTRIAL JUNO Tracker with tilt and activity detection and opening detection NB-IoT, LTE-CAT-M1 , GNSS, WIFI SSID Scan, with on board SIM, three batteries
S-JUNO-NBM1-TH-TRACK-2	JUNO IP67 TH Tracker, temperature and relative humidity tracker with tilt and activity detection and opening detection NB-IoT, LTE-CAT-M1 , GNSS, WIFI SSID Scan, with SIM card slot, two batteries
S-JUNO-NBM1- TH-TRACK-3	JUNO IP67 TH Tracker, temperature and relative humidity tracker with tilt and activity detection and opening detection NB-IoT, LTE-CAT-M1 , GNSS, WIFI SSID Scan, with on board SIM, three batteries
S-JUNO-IX-NBM1- TH-TRACK-2	INDUSTRIAL JUNO IP67 TH tracker, temperature and relative humidity tracker with tilt and activity detection and opening detection NB-IoT, LTE-CAT-M1 , GNSS, WIFI SSID scan, with SIM card slot, two batteries
S-JUNO-IX-NBM1- TH-TRACK-3	INDUSTRIAL JUNO IP67 TH Tracker, temperature and relative humidity with tilt and activity detection and opening detection NB-IoT, LTE-CAT-M1 , GNSS, WIFI SSID Scan, with on board SIM, three batteries



mioty



NB-IoT



GENERAL PROPERTIES

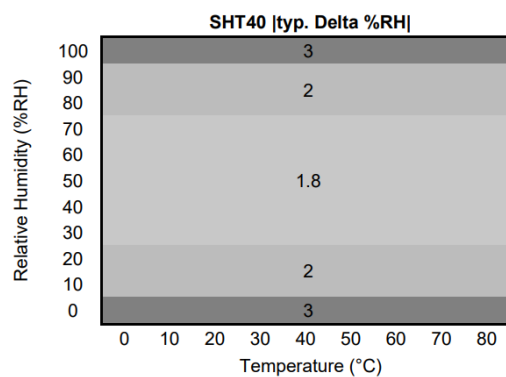
FEATURE	VALUE	UNIT
Dimensions lxbxh	86x86x25	mm
Weight	<200	g
Operating temperature	-25 to +75 (extended temperature range on request. Temperature range is limited by the maximum operating temperature of the primary cells)	°C
Maximum operating temperature Device	85	°C
Storage temperature (recommended)	15 to +30	°C
Rel. humidity during operation	5 to 99	%RH
IP Rating	• IP67/IP65 for -TH variants • IP69k for non-TH variants	
IK Rating	TBD	
Sensors	Temperature, relative humidity, acceleration sensor	
Traffic light function	no	
Audible alarm function	Yes, buzzer	
Provisioning	NFC	
Material	PBT GF20 + ASA	
NFC antenna	Integrated	
RF antenna	Integrated	
Accelerometer	Yes	
Batteries	Primary cell, replaceable, AA	
Primary cell type	LoRaWAN [®] , mioty [®] : • 2 x 3.6V LiSOCl2(recommended SAFT LS14500) Cellular: • 2 x 3V LiMnO2 spiral AA cell • 2 x 3.6V LiSOCl2 spiral AA cell	
Housing color	S-JUNO-XXXX black S-JUNO-IX-XXXX silk gray	
Assembly and installation	• Adhesive pad • Magnets • Screws • Gluing	

CONNECTIVITY AND RF SPECIFICATION

CHARACTERISTIC	WORTH	UNIT
RF Standards		MHz
Frequency	868	MHz
LoRaWAN®/mioty® EU		
Mobile radio frequency	Band 8, 20 (900 MHz, 800 MHz)	dBm
Transmission power	14	dBm
LoRaWAN®/mioty® EU		
Transmission power NB-IoT	23	dBm
LoRa® mac layer version	1.0.4	

SENSOR SPECIFICATION TEMPERATURE

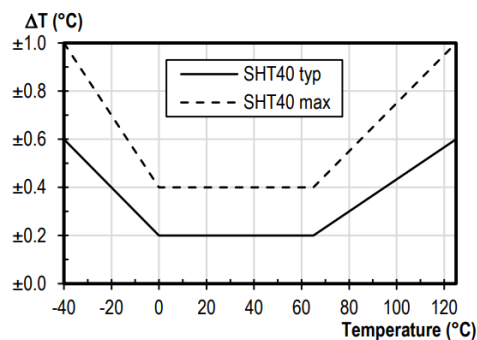
CHARACTERISTIC	WORTH	UNIT
Measuring range	-40 to +125	°C
Drift (long-term)	<0,03	°C/year
Accuracy	+/-0,2	°C
Resolution	0,1	°C
Repeatability	0,2	°C



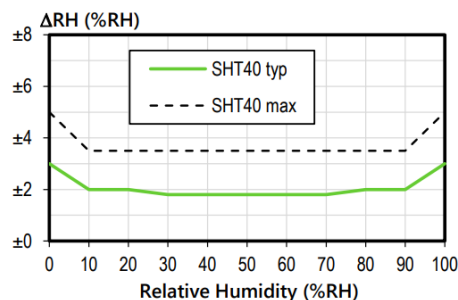
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SENSOR SPECIFICATION REL. HUMIDITY

CHARACTERISTIC	WORTH	UNIT
Measuring range	0 to +100	%RH
Drift (long-term)	<0,25	%RH/year
Accuracy	+/-1,8	%RH
Resolution	1	%RH
Repeatability	0,25	%RH



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HT_DS_Datasheet_SHT4x_5.pdf

More accurate temperature sensors available on request.

OVERVIEW OF TRACKING FEATURES

Item number	Radio standard	2.4 GHz scan	5 GHz scan	Maximum number of MAC addresses
S-JUNO(-iX)-LOEU-TRACK	LoRaWAN®	✓	X	6
S-JUNO(-iX)-LOEU-TH-TRACK	LoRaWAN®	✓	X	6
S-JUNO(-iX)-NBM1-TRACK-2	Cellular	✓	✓	20
S-JUNO(-iX)-NBM1-TRACK-3	Cellular	✓	✓	20
S-JUNO(-iX)-NBM1-TH-TRACK-2	Cellular	✓	✓	20
S-JUNO(-iX)-NBM1-TH-TRACK-3	Cellular	✓	✓	20
S-JUNO(-iX)-MIOTY-TRACK	mioty®	✓	✓	20
S-JUNO(-iX)-MIOTY-TH-TRACK	mioty®	✓	✓	20

LOCALIZATION LORAWAN SENSORS

CHARACTERISTIC	WORTH	UNIT
WIFI SSID Scan	Up to 6 MAC addresses on port 199	
WIFI SSID Scan Frequency	2,4	GHz
WIFI SSID scan accuracy	6 to 50	m
GNSS	GNSS scan with GPS and BeiDou	
GNSS accuracy	10 to 100	m
Antenna	Integrated	
Chipset	LR1110	
Feature for on-prem indoor tracking	Customization by configuring that up to 5 MAC addresses and RSSI values are transmitted in the standard uplink payload	
Tracking in the LoRaWAN	Network-dependent	
Tracking in Motion	Yes	

LOCALIZATION CELLULAR SENSORS

CHARACTERISTIC	WORTH	UNIT
WIFI SSID Scan	Up to 20 MAC addresses	
WIFI SSID Scan Frequency	2,4 and 5	GHz
WIFI SSID scan accuracy	2 to 50	m
GNSS	GPS, Galileo, GLONASS, BeiDou	
GNSS accuracy	5 to 100	m
Antenna	Integrated	
Cell Locate	Supported	
Tracking in Motion	Yes	

SENSOR SPECIFICATION INCLINATION

CHARACTERISTIC	WORTH	UNIT
Measuring range	0 to 360	°
Accuracy	+/-2°	°
Triggering in low power mode for opening detection	+/-60	°

SENSOR SPECIFICATION TEMPERATURE ACCELERATION SENSOR

CHARACTERISTIC	WORTH	UNIT
Measuring range	-40 to +85	°C
Accuracy	+/- 0,8	°C

TECHNICAL DRAWING

